NATIONAL HIGHER SCHOOL
OF MATHEMATICS
المدرســة الـوطنية
العليا في الرياضيات

1. (a) Compute the left, right, and midpoint Riemann sums for the function $f(x) = x^2$ over the interval $[0, 2]$ with $n = 4$ subintervals. Compare each approximation with the exact value of the integral.

(b) Evaluate the integral $\int_1^4 (3x + 1)\, dx$ using a right Riemann sum with $n = 6$. Verify your result by computing the exact integral.

(c) Use a midpoint Riemann sum with $n = 5$ to approximate the integral $\int_0^1 e^x\, dx$. Compare your approximation with the exact value $e - 1$.

(d) Set up the Riemann sum for the function $f(x) = \sin(x)$ over the interval $[0, \pi]$ using n subintervals and right endpoints. Express your answer in terms of n and do not evaluate the limit.

(e) Find the exact value of $\int_0^3 (2x + 1)\, dx$ by taking the limit of the Riemann sum as $n \to \infty$.

(f) Approximate the integral $\int_1^2 \frac{1}{x}\, dx$ using a left Riemann sum with $n = 4$. Compare your result with the exact value $\ln(2)$.

2. Use Riemann sums to find the following limits :

(a) $u_n = n \sum_{k=0}^{n-1} \frac{1}{k^2 + n^2}$

(b) $v_n = \prod_{k=1}^{n} \left(1 + \frac{k^2}{n^2}\right)^{\frac{1}{n}}$

(c) $\lim_{n\to\infty} \sum_{k=1}^{k=n} \frac{n}{4n^2+k^2}$

3. (a) Evaluate using symmetry :

$$\int_{-1}^{1} \frac{x^3}{\sqrt{1 - x^2}}\, dx$$

(b) Show that :

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx = \frac{\pi^2}{4}$$

(c) Compute using substitution :

$$\int_0^{\pi/2} \ln(\sin x)\, dx$$

(d) Show :

$$\int_0^1 \frac{\ln(1 + x)}{x}\, dx = \frac{\pi^2}{12}$$

4. Evaluate :

(a) $\displaystyle\int_0^1 \sqrt{-\ln x}\,dx, \quad \int_0^{\frac{\pi}{2}} \sin^n x\,dx$ for positive integers n.

(b) $\displaystyle\int_0^{\infty} x^2 e^{-x^2}\,dx, \quad \int_{-1}^{1} \frac{x^3+2x^2+3x+4}{(x^2+1)^2}\,dx, \quad \int_0^{\pi} e^{-x}\sin x\,dx.$

(c) $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{\sin x + \cos x}\,dx, \quad \int_0^1 (\ln x)^n\,dx$ for positive integers n

(d) $\displaystyle\int_0^{\infty} \frac{\ln x}{1+x^2}\,dx, \quad \int_0^1 x^a(1-x)^b\,dx$ for $a, b > -1$

5. (a) Evaluate :

$$\int_0^{\infty} \frac{x}{(x^2+1)^2}\,dx$$

(b) Show that :

$$\int_0^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

(Hint : Consider $\left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)^2$.)

6. Let

$$F(\alpha) = \int_0^1 x^{\alpha-1}\ln x\,dx$$

Show that $F(\alpha) = -\frac{1}{\alpha^2}$ for $\alpha > 0$.

7. Evaluate :

$$\int_0^{\infty} \frac{x^a}{e^x - 1}\,dx \quad \text{for } a > 0$$

8. Determine the nature of the following improper integrals

— $\int_0^{\infty} \frac{dx}{(1+e^x)(1+e^{-x})}, \quad \int_0^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx, \quad \int_0^1 \ln x dx, \quad \int_1^{\infty} \frac{\ln x}{x^2}dx$

— $\int_0^1 \frac{\ln x}{(1+x)^2}dx, \quad \int_0^1 \frac{1}{\sqrt{(1-x)(1+\alpha x)}}dx, \alpha \in \mathbb{R}$.

9. We want to calculate the value of the integral $I = \int_0^{\infty} \frac{\sin x}{x}dx$

(a) Show that this integral is convergent.

(b) Define the two following sequences $I_n = \int_0^{\pi/2} \frac{\sin((2n+1)x)}{\sin(x)}dx$, and $J_n = \int_0^{\pi/2} \frac{\sin((2n+1)x)}{x}dx$.

i) Using the change of variable $t = (2n+1)x$, show that $J_n \to I$.

ii) Show that $I_n - I_{n-1} = 0, \forall n \in \mathbb{N}^*$, and deduce that $I_n = \frac{\pi}{2}, \forall n \in \mathbb{N}$.

(a) Consider the function f defined on $(0, \pi/2)$ by $f(x) = \frac{1}{\sin x} - \frac{1}{x}$.

i) Show that this function extends continuously to zero into a C^1 function on $[0, \pi/2]$.

ii) Deduce that $\lim_{n\to+\infty}(I_n - J_n) = 0$ and that $I = \frac{\pi}{2}$.

10. Show that the following integrals are conditionally convergent :

$\int_{\pi}^{\infty} \frac{\cos x}{\sqrt{x}}dx\,; \quad \int_{-1}^{\infty} \cos(x^2)\,dx\ (\text{set } u = x^2)\,; \quad \int_{\pi}^{\infty} x^2\sin(x^4)\,dx.$